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THE RATTLESNAKE HILLS OF CENTRAL WYOMING REVISITED: FURTHER PALEOMAGNETIC RESULTS

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Abstract. Paleomagnetic results from the Eocene igneous rocks of the Rattlesnake Hills in central Wyoming have added 9 reliable virtual geomagnetic pole positions to previous paleomagnetic data from the area. These new data when combined with corrected results from the previous study yield a paleomagnetic pole located at 146.2° E, 79.4° N; α_{95} = 9.6°. This pole position, at about 44 mybp, agrees very well with other Eocene paleopole determinations for stable North America. The proximity of these pole positions to the present rotation axis shows that most of the angular distance between the apparent pole position for the Late Cretaceous and the present was closed in the Paleocene and/or early Eocene.

Introduction

The Rattlesnake Hills of central Wyoming consist of a northwest-trending Laramide anticline bordered by the Granite Mountains on the south and the Wind River Basin on the north. Flanking the Precambrian metamorphic core of the anticline are Paleozoic and Mesozoic sedimentary rocks which, in turn, are partially covered by flat-lying Eocene sedimentary and pyroclastic deposits (Figure 1). As discussed by Pekarek et al. [1974] more than 40 intrusive bodies and volcanic vents consisting of rhyolite, quartz latite, latite, trachyte, phonolite, and vogesite crosscut the Precambrian through early Tertiary rocks. These Eocene igneous rocks cover about 400 km^2; individual igneous bodies vary in diameter from 75 m to 2500 m.

The history of volcanism in the Rattlesnake Hills, summarized by Pekarek et al. [1974], begins in Middle Eocene time with rhyodacite volcanism followed by alkalic volcanism; both types continued concurrently until cessation of volcanic activity in Late Eocene time. Radiometric age dates (corrected for new IUGS constants [Dalrymple, 1979]) associated with Rattlesnake Hills volcanism include: 1) a K-Ar date of 50.3 mybp by Evernden et al. [1964] from a unit just below the Wagon Bed Formation, which contains volcanic debris from Rattlesnake Hills volcanism [Van Houten, 1964], 72 km southwest of the Rattlesnake Hills and 2) two K-Ar dates (45.1 $\pm$ 2.6 mybp and 44.7 $\pm$ 1.0 mybp) by Pekarek et al. [1974] from quartz latite and phonolite respectively. These latter two samples were selected in an attempt to date the youngest volcanic activity. However, rhyolites discovered after sampling for the radiometric dating are most likely younger than the reported dates [Pekarek et al., 1974]. Consequently radiometric information suggests that duration of volcanism in the Rattlesnake Hills was on the order of 8 million years or less.

Lack of extensive spatial overlapping of the igneous events prevents determination of the sequence of volcanism from field relationships. The high reversal frequency of the geomagnetic field during the Eocene suggested to Shive et al. [1977] that paleomagnetic directions from the igneous units might provide information allowing dating or grouping of volcanic episodes. Shive et al. [1977] also recognized the importance of an addition to the reliable Eocene paleomagnetic pole positions for stable North America.

The results of Shive et al. [1977] were different than expected in that: 1) none of the 10 igneous bodies they studied are normally magnetized and 2) when plotted on a stereonet, virtual geomagnetic pole positions (VGP) from the bodies were distributed along a small circle. This led them to suggest that: 1) duration of volcanic activity in the Rattlesnake Hills was 1.7 m.y. or less and 2) the small circle progression of VGP positions was caused by an excursion which, when correlated with lithologies, may imply an igneous history of vogesite followed by quartz latites then phonolites. Thinking that perhaps their small circle was the result of normal paleosecular variation rather than the result of an excursion, Shive et al. [1977] also published a tentative paleomagnetic pole for the Rattlesnake Hills. Obviously, more paleomagnetic work was necessary in order to test these hypotheses; the purpose of this paper is to summarize the paleomagnetic work which is now complete.

Sampling and Laboratory Methods

We collected oriented samples from 12 separate igneous bodies representing compositions of quartz latite, rhyolite, phonolite, latite, and vogesite. Figure 1 shows the location of the sampling sites; lithologies are listed in Table 1. Combining these new sites with those of Shive et al. [1977] gives us a total of 22 paleomagnetic sites from the Eocene volcanic rocks of the Rattlesnake Hills. This represents about half of the 40 or so intrusive bodies and volcanic vents in the area. Because some of the igneous events are clearly synchronous [Pekarek, 1974] we feel that our sampling density has been more than adequate to characterize the paleomagnetic nature of the Tertiary igneous rocks in the Rattlesnake Hills. Rock drills born of chain saws were used to collect between 6 and 10 samples per site. Samples were oriented *in situ* with a magnetic compass; back azimuths were taken to insure against declination error resulting from local magnetic anomalies.

Remanent magnetization was measured with a Schonstedt SSM-1 spinner magnetometer. Three specimens from each site were selected as pilot specimens; these were subjected to stepwise alternating field demagnetization. Peak fields for the pilot specimens were 50, 100, 200, and 400 oersted. The initial demagnetization level for each site was chosen as that field which minimized the angular dispersion of the pilot

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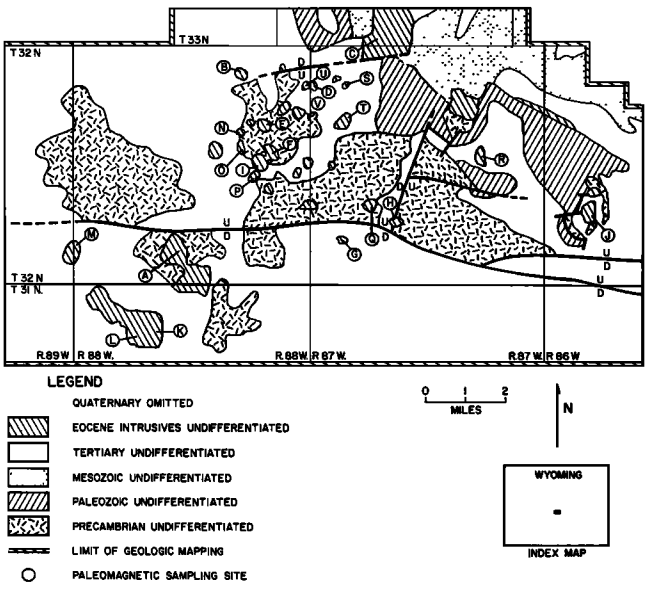


Figure 1. Index map of the Rattlesnake Hills, central Wyoming.

specimens from that site. Subsequently, most sites were demagnetized in higher fields with all specimens from the site being demagnetized at each successive interval. The optimum demagnetization level for a site was chosen as the level which maximized K (best estimate of Fisher's [1953] precision parameter). No consistent secondary components of magnetization were observed. The alternating field demagnetization was performed, over a period of many months, with two different demagnetizing systems: 1) a "four-axis" tumbler system and 2) a Schonstedt GSD-1, non-tumbling, specimen demagnetizer. When tested for consistency, both units gave equivalent results at comparable demagnetization levels even after storage tests of several months.

Most specimens responded well to alternating field demagnetization tests. Essentially all specimens used for statistical analysis had attained a stable end point at the optimum demagnetization field. Three sites were excluded from statistical analysis because of poor grouping at all demagnetization levels; for all three, K was less than 5. Although these three sites were deemed unreliable for inclusion in the angular dispersion or pole position calculations, their behavior during the determination of magnetic stability suggested that they were magnetized during a reversed configuration of the geomagnetic field. In addition to the reliability criteria discussed above, individual specimens that remained at more than twice the angular standard deviation from their respective site mean directions at the optimum demagnetization level were excluded from further statistical analysis. This criteria affected 6 specimens from this study and 7 from that of Shive et al. [1977].

No tectonic corrections were necessary. The largest post-igneous fault movement was of a translational nature and less than 100 m of offset occurred [Pekarek, 1974]. All field indications are that the Eocene igneous rocks have remained as formed with negligible between-site rotations.

Magnetic Mineralogy

In order to determine the dominent magnetic minerals of the Rattlesnake Hills volcanic rocks,

Table 1: N, number of samples; H_o, optimum demagnetizing field in oersteds; Dec. and Inc, mean declination and inclination of site; α_{95} and K, Fisher statistical parameters; δ, angular standard deviation; Plat and Plong, latitude (north positive) and longitude (east positive) of VGP (or mean pole); #, sites not included in calculation of mean.

SITE	LITHOLOGY	N	H_o	DEC	INC	α_{95}	δ	K	PLAT	PLONG
A	Quartz Latite	7	100	187.6	-58.5	5.2	6.5	135.1	83.2	12.2
B	Phonolite	6	200	189.4	-62.3	5.9	6.5	131.4	83.1	332.8
C	Quartz Lttite	9	800	202.0	-57.7	8.2	12.1	40.3	72.8	350.2
D	Vogesite	6	200	159.4	-20.2	11.0	12.1	37.8	53.0	107.0
E	Phonolite	7	400	180.9	-58.0	10.0	12.3	37.6	85.8	63.0
F	Trachyte#	8	800	98.0	-8.8	---	74.3	<3.0	8.8	160.5
G	Phonolite	6	400	148.8	-75.9	26.5	27.6	7.6	63.0	222.8
H	Quartz Latite	8	200	149.0	-71.1	2.4	3.2	548.6	66.3	206.2
I	Latite	8	200	134.7	-62.7	7.9	10.7	50.0	57.5	179.2
J	Quartz Latite	9	400	144.0	-65.4	5.8	8.6	80.0	64.4	185.9
K	Quartz Latite#	10	1000	135.3	-63.4	---	60.2	<3.0	58.1	182.4
L	Quartz Latite	5	1000	178.0	-71.7	21.8	20.0	13.3	76.2	247.9
M	Rhyolite	8	400	178.5	-47.1	3.6	4.9	237.9	75.5	77.8
N	Phonolite	5	400	181.7	-61.1	25.0	22.7	10.3	88.6	8.4
O	Phonolite#	4	100	183.3	-21.8	50.0	34.1	4.4	58.4	66.5
P	Latite	7	800	165.0	-16.0	25.4	29.5	6.6	53.1	97.9
Q	Phonolite	5	400	174.2	-60.7	13.7	12.8	32.2	85.6	151.6
R	Rhyolite	7	400	137.5	-67.7	4.5	5.6	179.6	60.3	193.2
S	Phonolite	6	400	160.8	-58.9	14.2	15.3	23.5	75.2	156.9
T	Phonolite	6	600	148.8	-29.7	15.5	16.7	19.7	52.2	127.1
U	Vogesite	9	400	169.5	-20.9	15.4	22.1	12.1	56.8	91.8
V	Phonolite#	6	800	130.3	-33.7	---	54.1	<3.0	41.7	148.3
Mean Paleomagnetic Pole		18				9.6	21.3	13.9	79.4	146.2

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Curie temperatures for several representative samples were determined by thermomagnetic analysis. The analyses were performed in vacuum ($\sim 10^{-4}$ torr) and in fields of 3 to 8 kilogauss. In all cases except for phonolites reliable ferrimagnetic signals were obtained from the samples. In the case of phonolites we could not determine Curie temperatures of the chosen samples because a paramagnetic signal overwhelmed the signal from any ferrimagnetic components. This behavior occurred even though the NRM intensity of the two phonolite samples was on the order of $6X10^{-2}$ emu/cm^3. Curie temperature determinations (Figure 2) from the remaining rock types suggest that low-titanium titanomagnetites are the dominant magnetic phase of these rocks. The irreversible curve of sample 77008 results from the inversion of an oxidized low-titanium titanomagnetite. The irreversible curves of samples 77021 and 77080 may result from a similar inversion or from slight oxidation during the heating stage of the thermomagnetic analysis.

Summary of Paleomagnetic Results

Table 1 summarizes the available paleomagnetic data for the Rattlesnake Hills. Sites A through J represent data corrected from Shive et al. [1977]. Sites K through V are new results from this investigation. In the data of Shive et al. [1977] all but one of the VGP positions were incorrectly calculated. In addition, their 95% circles of confidence were underestimated by up to 20%. Because of the error in their VGP locations their results and conclusions are not valid.

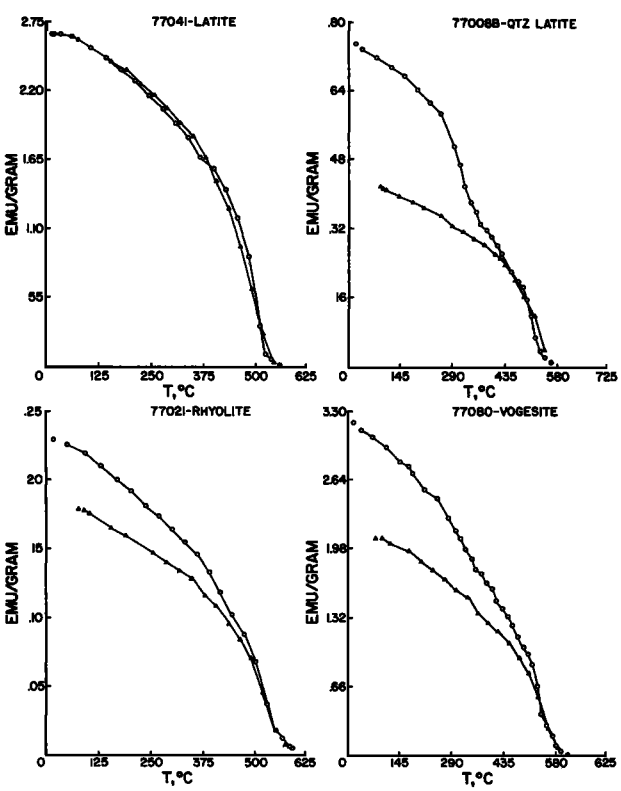


Figure 2. Results of thermomagnetic analyses for 4 representative samples of the Rattlesnake Hills volcanic rocks. Circles are heating curves; triangles represent cooling.

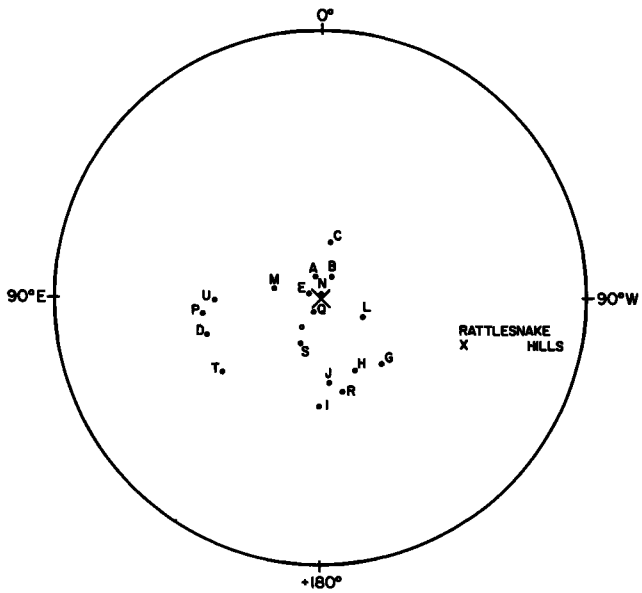


Figure 3. Equal-area plot of the 18 VGP's considered in statistical analyses and paleomagnetic pole position calculation. The open circle represents the mean position, the small cross marks the sampling location.

Observation of Table 1 shows that 18 of the 22 sites are clearly reversely magnetized. Sites F, K, O, and V, although not considered reliable for statistical purposes, also represent a reversed configuration of the geomagnetic field. It is of note that the angular standard deviation for sites A through J is somewhat smaller than for sites K through V. We cannot definitely explain this but we suspect that it may result from sampling the "best" outcrops during the first investigation.

VGP positions for the 18 acceptable sites are shown in Figure 3. Ovals of 95% confidence for the VGP's have been omitted from Figure 3 for clarity. In contrast to the distribution of VGP positions presented by Shive et al. 1977 the azimuthal distribution of VGP positions is reasonably uniform. None of the VGP's are greater than 30° from the mean position. All 18 VGP's represent a stable reversed state of the geomagnetic field.

The paleomagnetic pole for the Rattlesnake Hills Eocene igneous rocks, calculated as the mean of the 18 acceptable VGP's, lies at longitude 146.2° E, latitude 79.4° N. Using the statistical techniques of Fisher [1953] gives α_{95} = 9.6° and K = 13.9 for the above pole. The angular standard deviation of the VGP positions about their mean is 21.3°. This term, δ_T, consists of two components: 1) the paleofield dispersion, δ_F and, 2) the within-site dispersion, δ_W; $\delta_T^2 = \delta_F^2 + (\delta_W^2/\bar{n})$ [McElhinny and Merrill, 1975], where $_W$ is the average dispersion at each of N sites and $\bar{n}$ is the average number of samples per site. The field dispersion for the Eocene volcanic rocks of the Rattlesnake Hills is 20.7°. This value falls within the limits of paleosecular variation model E of Baag and Helsley [1974].

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Discussion

The results of this study, combined with the corrected results of Shive et al. [1977], necessitates considerable revision regarding aspects of Rattlesnake Hills volcanism inferred from the paleomagnetic data. Although the previous study suggested that volcanism had sampled an excursion of the geomagnetic field, the VGP distribution of all 18 acceptable sites (Figure 3) shows no hint of anomalous geomagnetic field behavior. The distribution of VGP's (Figure 3) and the associated value for paleofield dispersion of 20.7° shows that duration of volcanism was great enough to average secular variation of the magnetic field to a reasonable value. Hence, we feel that: 1) many of our sites represent independent measurements of the Eocene geomagnetic field and, 2) volcanism did not occur during one very short interval. Although all of the sites for the Rattlesnake Hills are reversed polarity (Table 1) the azimuthal distribution of the VGP's about their mean is uniform (Figure 3). This relation further decreases the likelihood that volcanic activity was confined to a single short interval of time. The possibility that volcanism occurred during two or more reversed intervals is only slightly less likely than that of volcanism occurring during a single reversed interval.

The radiometric dates of 44.7 mybp and 45.1 mybp from Pekarek et al. [1974] are associated directly with volcanic units we have sampled. In contrast, the 50.3 mybp date of Evernden et al. [1964] represents the upper age limit on Rattlesnake Hills igneous activity. Because of this we feel that volcanism is most likely to have occurred during one or two reversed intervals near 44 mybp. The geomagnetic polarity time scale of Ness et al. [1979] shows a normal interval between 43.7 and 45.2 mybp that corresponds to anomaly 20. Volcanism could have occurred during any of several reversed intervals near anomaly 20.

The pole position for the Rattlesnake Hills (146.2° E, 79.4° N, α_{95} = 9.6°) is west of, but at the 95% confidence level indistinguishable from, the average Late Eocene paleopole for stable North America (164° W, 83° N) of Irving [1979]. Paleopole positions determined from two Early Eocene igneous centers in north-central Montana [Diehl et al., 1980] agree very well with the above results. The coincidence of these recent additions to the Eocene paleomagnetic record with the average Late Eocene paleopole of Irving [1979] shows that the latter is a very good Eocene reference pole for stable North America. Although these pole positions are distinct from the present day rotation axis, their proximity to it shows that most of the angular distance between the apparent pole position for the Late Cretaceous and the present rotation axis was closed in the Paleocene and/or Early Eocene.

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